Computer Networks

PPT-4: OSI Model

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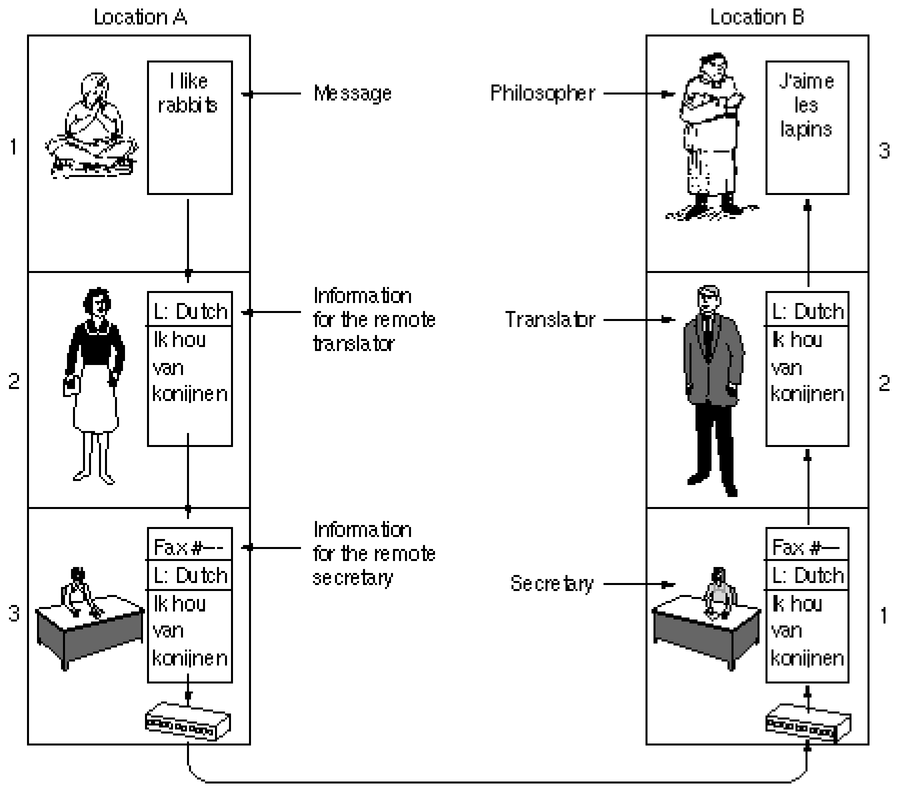
Network Software

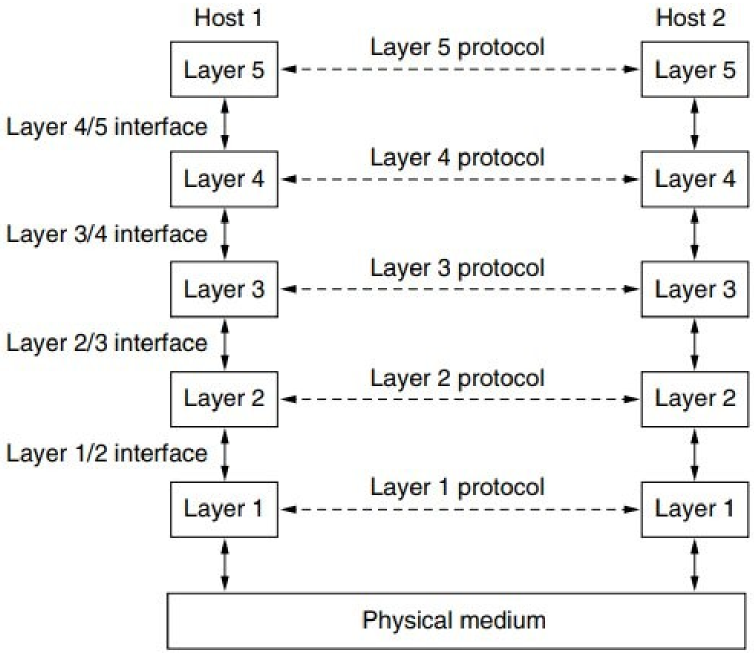


LAYERING: The philosopher-translator-secretory architecture

Layer, protocols & interfaces

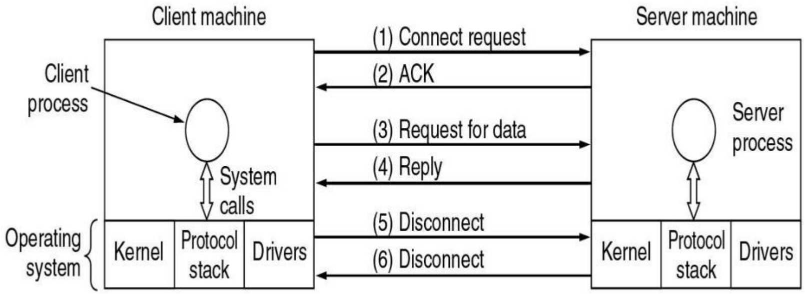
- Concept/architecture
- Addressing
- Error Control
- Flow Control
- Multiplexing/Demultiplexing
- Routing

- Connection-Oriented/Connectionless
- Quality of Service (QOS)
- byte-stream/message sequences
- Datagram / Acknowledged Datagram service
- Request-reply service



Service Primitives

Primitive	Meaning
LISTEN	Block waiting for an incoming connection
CONNECT	Establish a connection with a waiting peer
RECEIVE	Block waiting for an incoming message
SEND	Send a message to the peer
DISCONNECT	Terminate a connection



Client - Server interaction on a connection oriented network

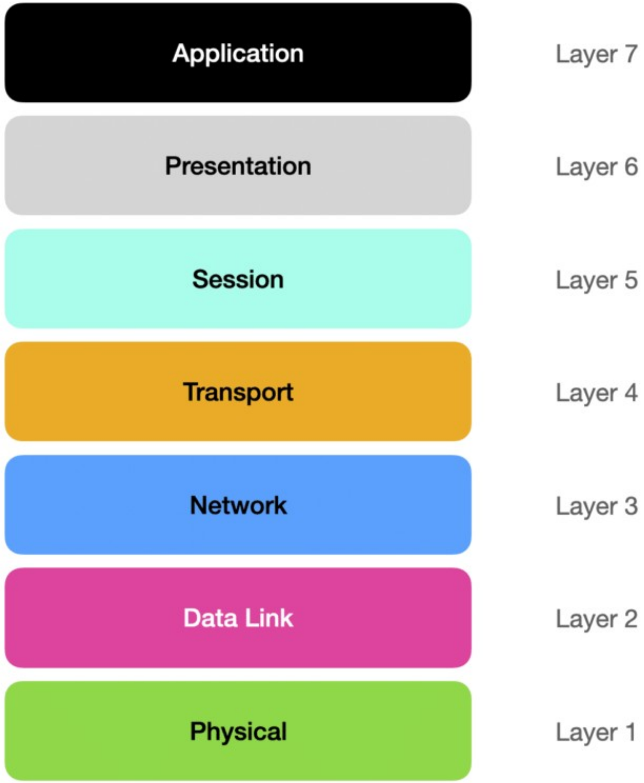
OSI Model

The OSI (Open Systems Interconnection) model is a conceptual framework that **standardizes** the functions of a telecommunication or computing system into **seven abstraction layers**. It was developed by the International Organization for Standardization (**ISO**) to facilitate communication and **interoperability between different systems** and devices. **Each layer in the OSI model has a specific set of functions and interacts with adjacent layers, providing a structured approach to understanding and designing network architectures**.

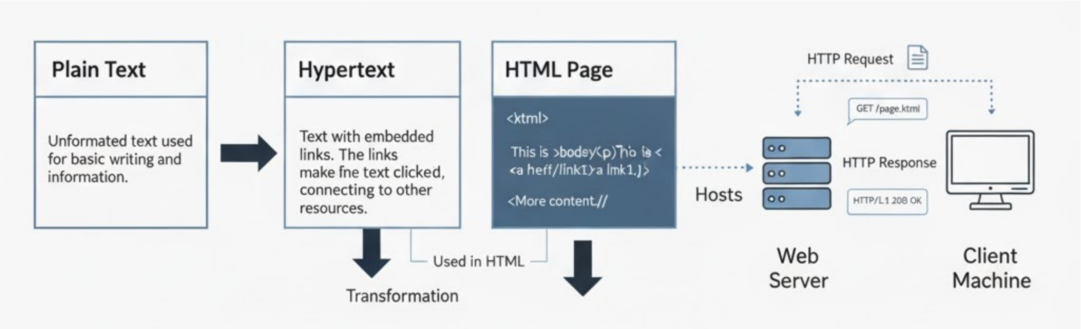
OSI MODEL

Protocol Stack: Protocol is an agreement between the communication parties on how communication is to proceed.

- Abstraction
- Modularity
- Interoperability
- Ease of Understanding

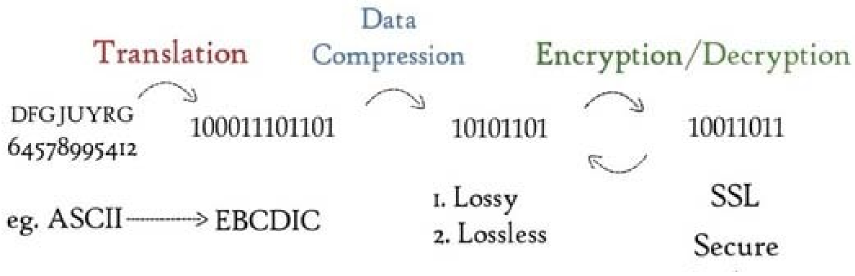


Application Layer

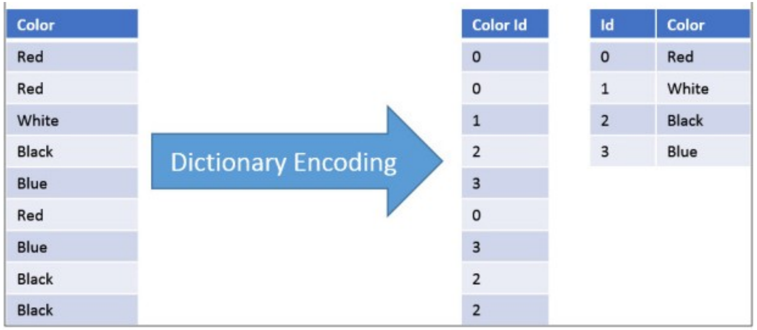


- Top layer of the OSI model
- Provides network services to end-user applications
- Acts as an interface between user and network
- Enables services like web browsing, email, and file transfer
- Uses protocols like HTTP

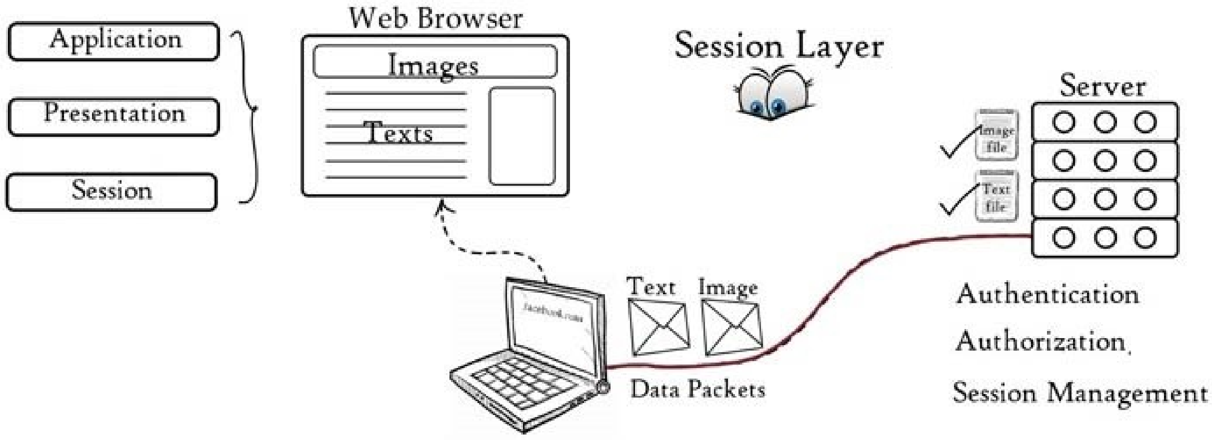
Presentation layer



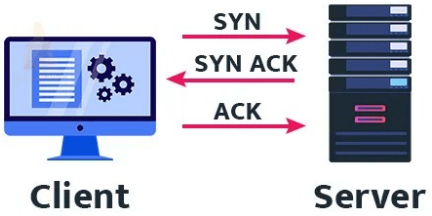
- Responsible for data formatting and translation
- Encrypts and decrypts data
- Performs data compression and decompression
- Ensures data is in a readable format for the application layer

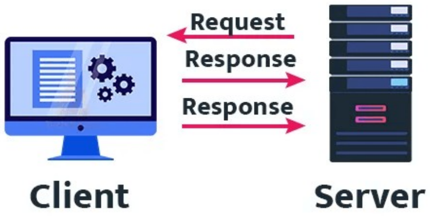


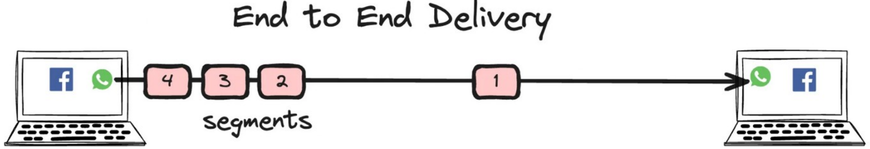
Session layer

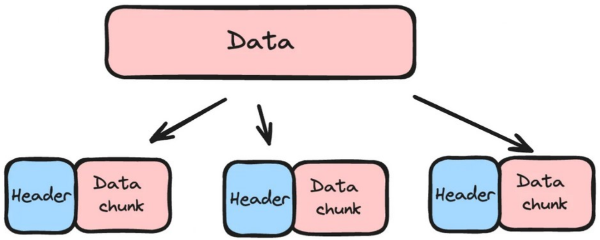


- Establishes, manages, and terminates sessions between devices
- Controls dialog between sender and receiver
- Provides synchronization using checkpoints
- Handles session recovery in case of interruption



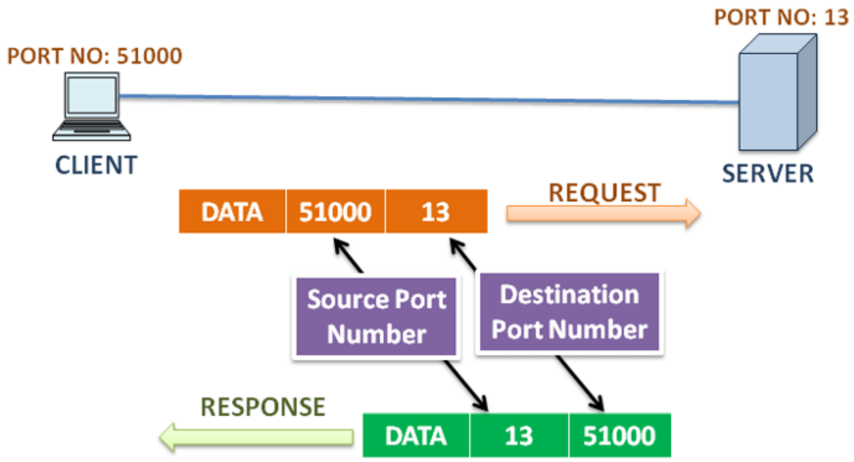




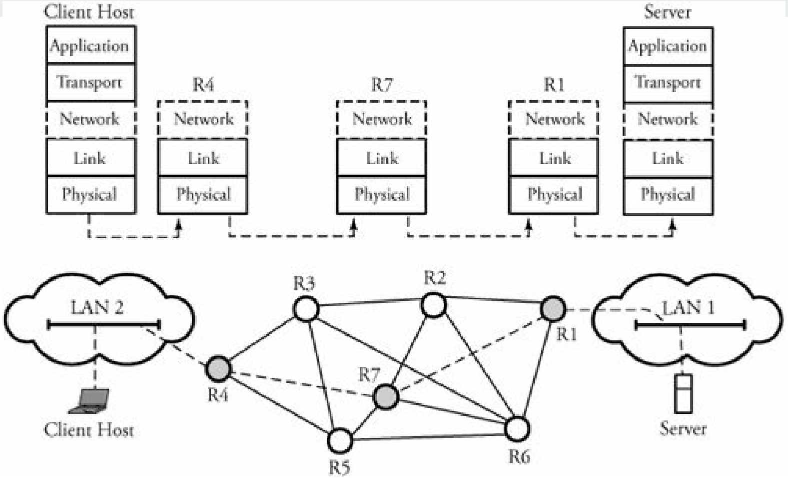


Transport Layer

- Provides end-to-end communication between devices
- Segments and reassembles data
- Provides end-to-end, port-to-port communication between processes, ensuring data reaches the correct application.
- Uses protocols like TCP (reliable) and UDP (faster, connectionless)



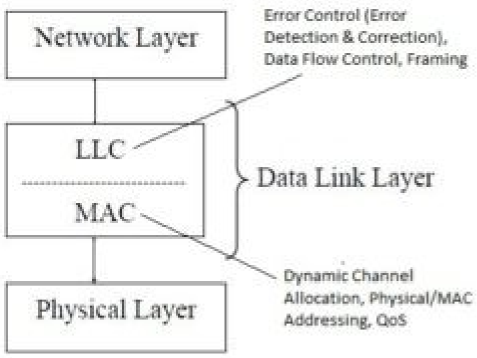
Network Layer



- Handles logical addressing using IP addresses
- Determines the best path for data delivery (routing)
- Responsible for packet forwarding between devices across networks
- Manages fragmentation and reassembly of packets

Splits segments (from Transport Layer) into smaller packets to fit network constraints like Maximum Transmission Unit (MTU)

Simple Analogy
Segmentation → Cutting a book into chapters for easier handling (by the receiver)
Fragmentation → Cutting chapters into pages that can fit through a narrow tube (network links)



Data link Layer

- Provides node-to-node (link-to-link) data transfer
- Detects and may correct errors from the Physical Layer
- Handles framing of data for transmission

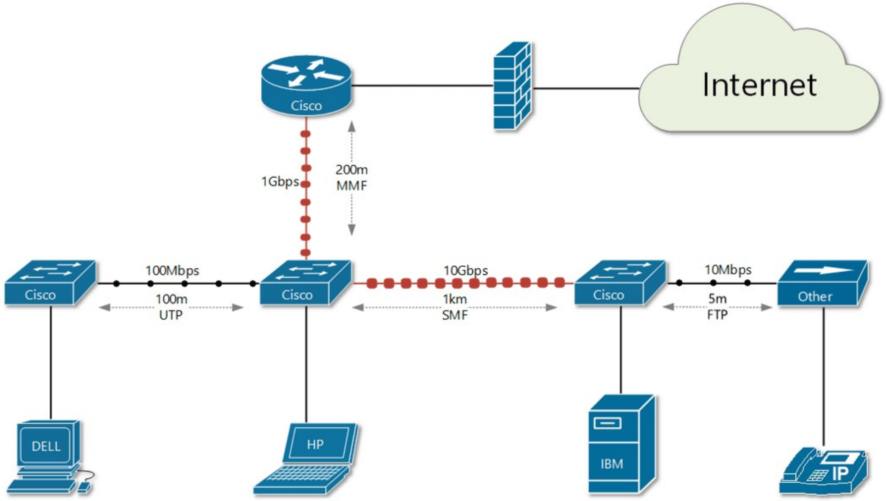
Sublayers of Data Link Layer

1. Logical Link Control (LLC)
    - Manages error checking and flow control
    - Identifies which network layer protocol is being used
2. Media Access Control (MAC)
    - Controls how devices access the physical medium
    - Determines who can send data and when
    - Handles physical addressing (MAC addresses)

In a cellular network, multiple users in a cell share frequency bands. When a user makes a call or sends data, the system dynamically allocates an available channel. After use, the channel returns to the pool for others.

Physical Layer

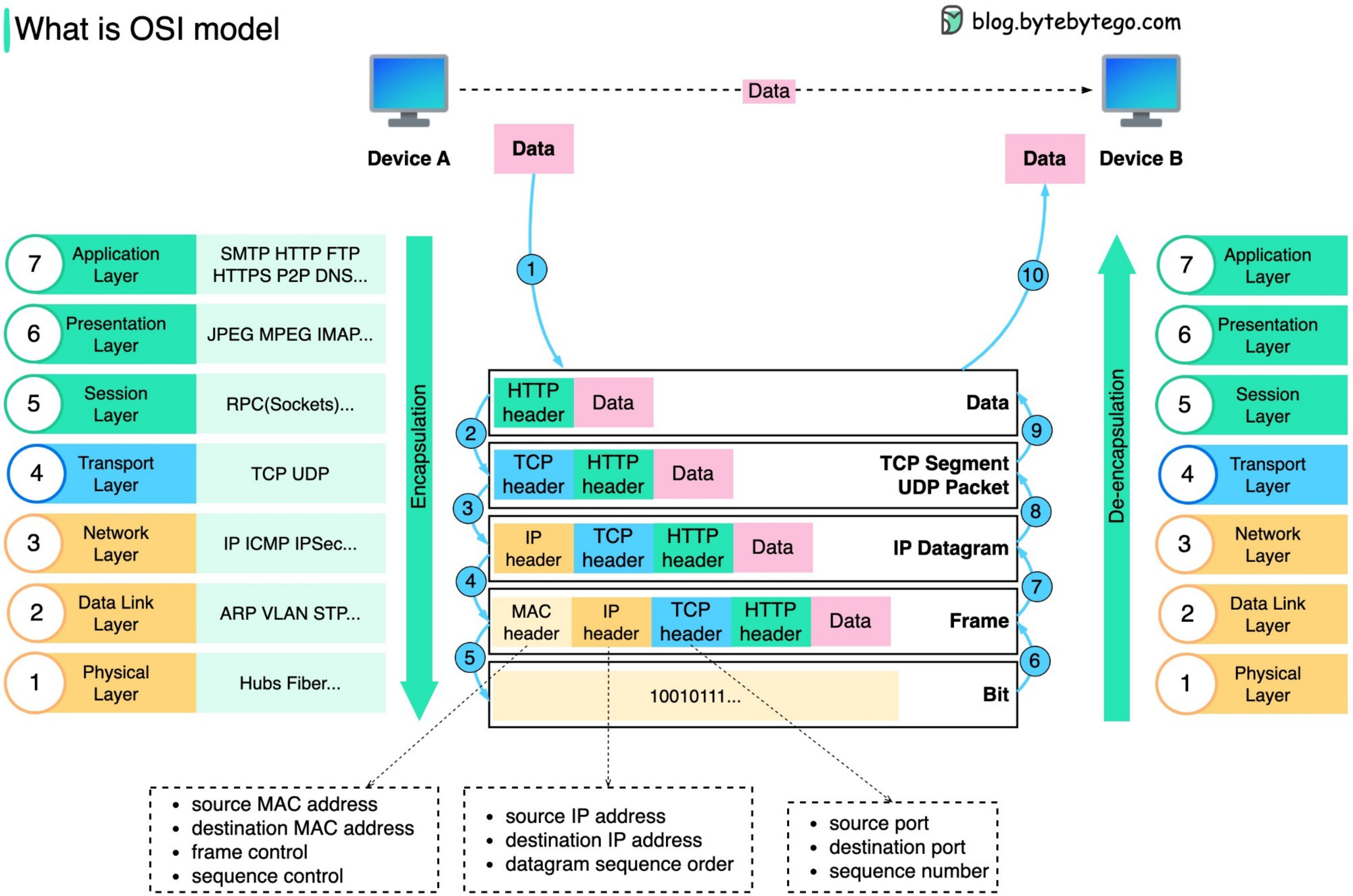
- Deals with physical transmission of raw bits over a medium
- Defines hardware specifications like cables, connectors, and wireless signals
- Handles bit rate, voltage levels, and modulation
- Responsible for converting data into signals for transmission and reception
- Includes devices like hubs, repeaters, and network interface cards (NICs)



OSI Model

End User Application Layer: HTTP, DNS, FTP, SSH	**Application**	Layer 7
Syntax and Semantics/ Translation, Encryption and Compression: SSL, MPEG , JPEG, PNG	**Presentation**	Layer 6
Session, dialog control, token management, synchronisation: checkpointing long transmission: resume after crash	**Session**	Layer 5
Process to Process / Port to Port: Segmentation / Reassembly, Connection control, end to end flow control, error control	**Transport**	Layer 4
Logical Addressing : IP addresses, Routing of packets from source to destination network, congestion control, QoS (delay transit time and jitter), heterogeneous network	**Network**	Layer 3
Framing, flow control fast/slow transmitter, Error Control, buffer management, Physical Addressing: Media Access Control for shared medium/ channel	**Data Link**	Layer 2
Transmitting raw bits over a communication channel, voltage, pins in connector, mechanical and electrical and timing interfaces & physical transmission medium	**Physical**	Layer 1

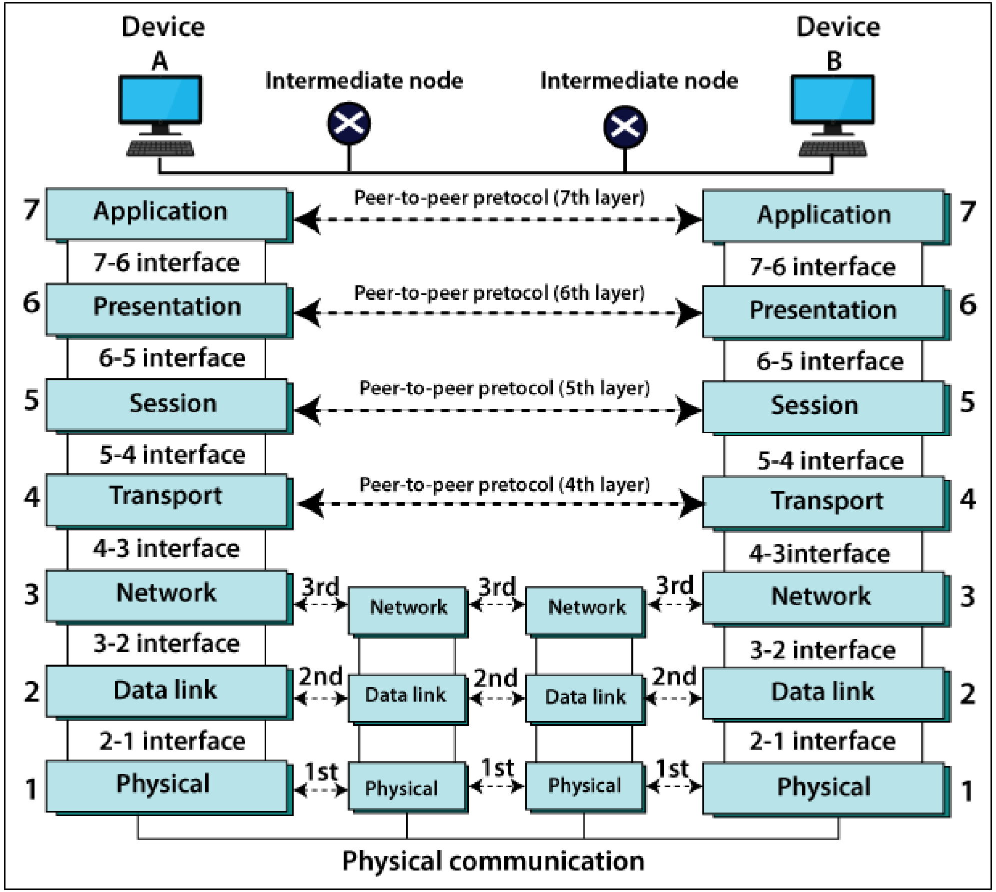
What is OSI model



Intermediary Nodes

- Network
- DLL
- Physical

- Repeater
- HUB
- Bridge
- Switch
- Router



Next Lecture

TCP/IP Model